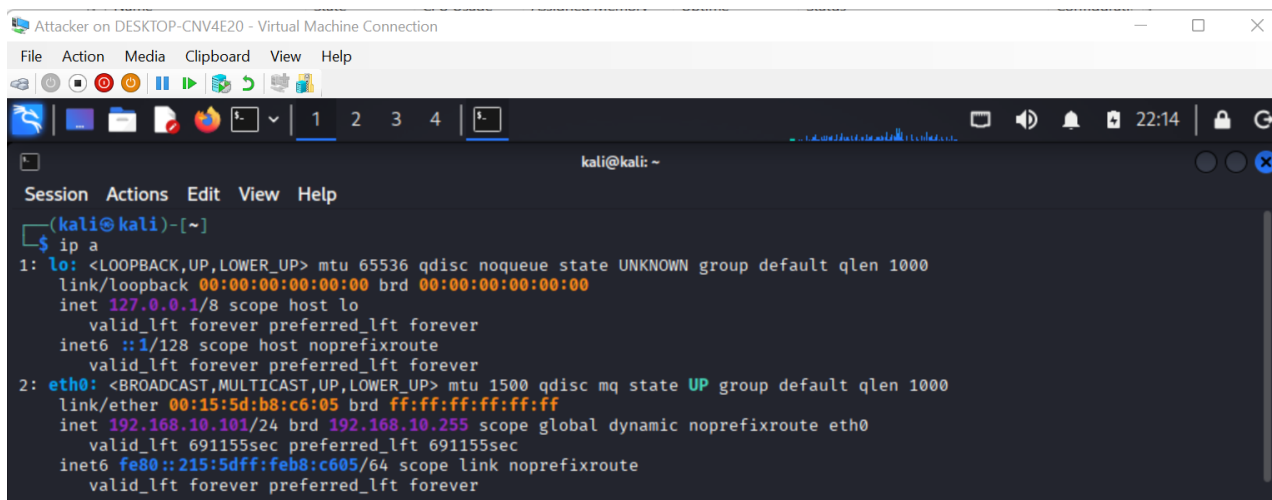


Kali Linux Connected to the Internal Network

Description: Kali Linux configured to communicate with the Active Directory lab through the Hyper-V Internal Virtual Switch.

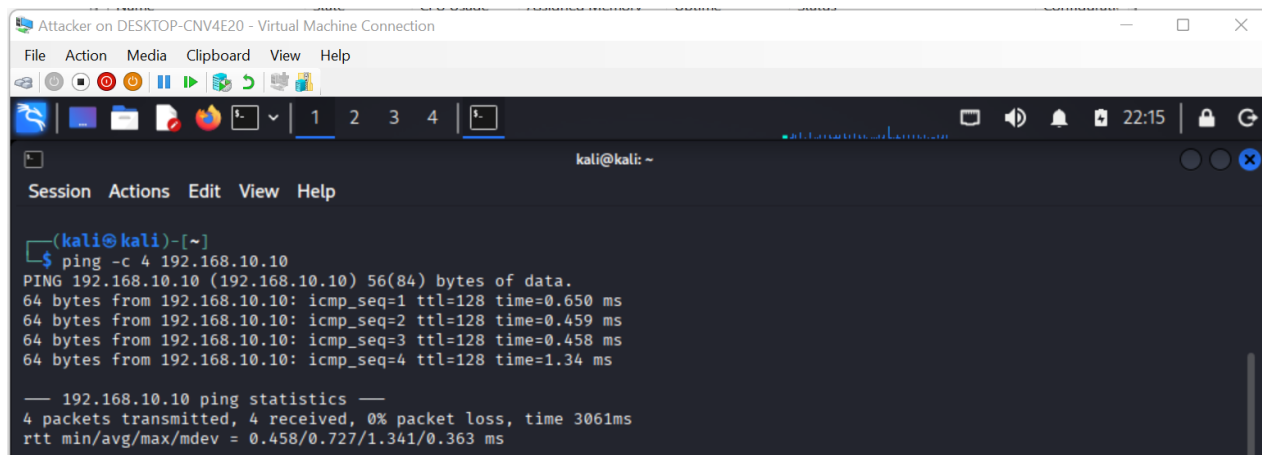


The screenshot shows a Kali Linux terminal window titled "Attacker on DESKTOP-CNV4E20 - Virtual Machine Connection". The terminal displays the output of the 'ip a' command, showing the configuration for the loopback interface 'lo' and the ethernet interface 'eth0'. The 'lo' interface has an IP address of 127.0.0.1. The 'eth0' interface has an IP address of 192.168.10.101 and a MAC address of 00:15:5d:b8:c6:05.

```
(kali@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:b8:c6:05 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.101/24 brd 192.168.10.255 scope global dynamic noprefixroute eth0
        valid_lft 691155sec preferred_lft 691155sec
    inet6 fe80::215:5dff:feb8:c605/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Network Connectivity Test

Description: Successful connectivity test between Kali Linux and the Domain Controller.



The screenshot shows a Kali Linux terminal window titled "Attacker on DESKTOP-CNV4E20 - Virtual Machine Connection". The terminal displays the output of the 'ping -c 4 192.168.10.10' command, showing four successful ping requests to the IP address 192.168.10.10. The ping statistics show 4 packets transmitted, 4 received, 0% packet loss, and a time of 3061ms.

```
(kali@kali)-[~]
$ ping -c 4 192.168.10.10
PING 192.168.10.10 (192.168.10.10) 56(84) bytes of data:
64 bytes from 192.168.10.10: icmp_seq=1 ttl=128 time=0.650 ms
64 bytes from 192.168.10.10: icmp_seq=2 ttl=128 time=0.459 ms
64 bytes from 192.168.10.10: icmp_seq=3 ttl=128 time=0.458 ms
64 bytes from 192.168.10.10: icmp_seq=4 ttl=128 time=1.34 ms

--- 192.168.10.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3061ms
rtt min/avg/max/mdev = 0.458/0.727/1.341/0.363 ms
```

Nmap Service Discovery

Description: Nmap scan results showing the open ports and services running on the Domain Controller.

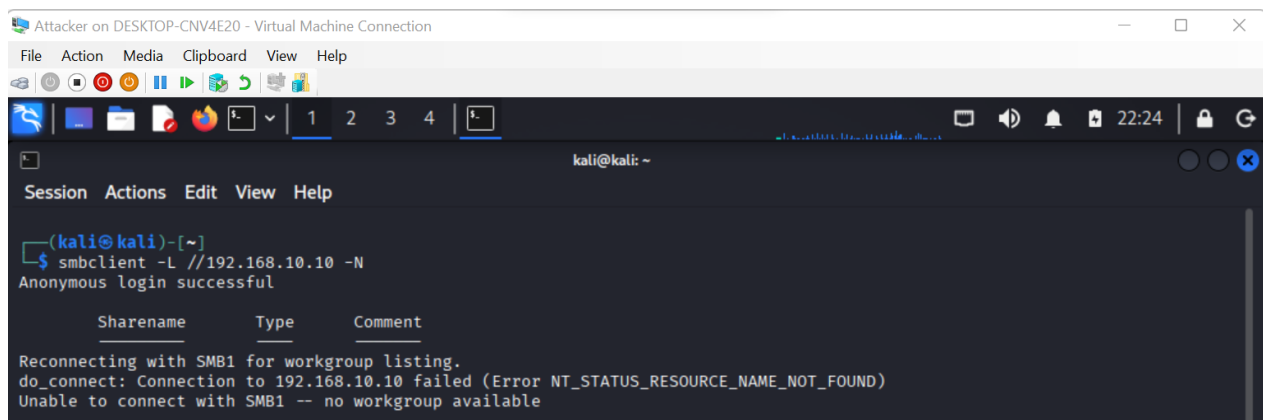
```
(kali@kali)-[~]
$ nmap -sn 192.168.10.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-27 22:06 +0100
Nmap scan report for 192.168.10.1
Host is up (0.00036s latency).
MAC Address: 00:15:5D:B8:C6:09 (Microsoft)
Nmap scan report for 192.168.10.10
Host is up (0.00065s latency).
MAC Address: 00:15:5D:B8:C6:03 (Microsoft)
Nmap scan report for 192.168.10.100
Host is up (0.0011s latency).
MAC Address: 00:15:5D:B8:C6:0A (Microsoft)
Nmap scan report for 192.168.10.101
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 10.91 seconds
```

```
Attacker on DESKTOP-CNV4E20 - Virtual Machine Connection
File Action Media Clipboard View Help

kali@kali: ~
Session Actions Edit View Help
|_nbtstat: NetBIOS name: DC01, NetBIOS user: <unknown>, NetBIOS MAC: 00:15:5d:b8:c6:03 (Microsoft)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 60.75 seconds

(kali@kali)-[~]
$ sudo nmap -O 192.168.10.10
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-27 22:13 +0100
Nmap scan report for 192.168.10.10
Host is up (0.00082s latency).
Not shown: 987 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
88/tcp    open  kerberos-sec
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
389/tcp   open  ldap
445/tcp   open  microsoft-ds
464/tcp   open  kpasswd5
593/tcp   open  http-rpc-epmap
636/tcp   open  ldapssl
3268/tcp  open  globalcatLDAP
3269/tcp  open  globalcatLDAPssl
5357/tcp  open  wsdapi
5985/tcp  open  wsman
MAC Address: 00:15:5D:B8:C6:03 (Microsoft)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
Running (JUST GUESSING): Microsoft Windows 2022|11|2016 (97%)
OS CPE: cpe:/o:microsoft:windows_server_2022 cpe:/o:microsoft:windows_11 cpe:/o:microsoft:windows_server_2016
Aggressive OS guesses: Microsoft Windows Server 2022 (97%), Microsoft Windows 11 21H2 (91%), Microsoft Windows Server 2016
```

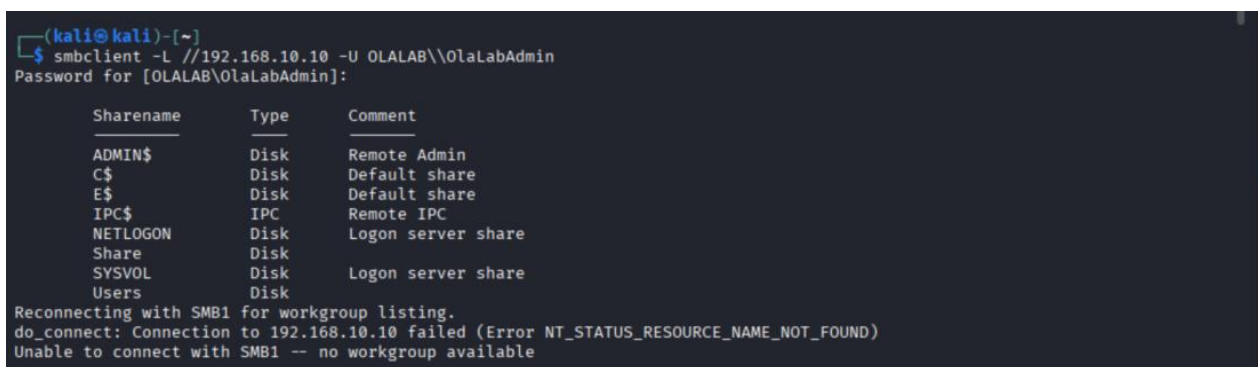


```
(kali@kali)-[~]
$ smbclient -L //192.168.10.10 -N
Anonymous login successful

      Sharename      Type      Comment
-----
Reconnecting with SMB1 for workgroup listing.
do_connect: Connection to 192.168.10.10 failed (Error NT_STATUS_RESOURCE_NAME_NOT_FOUND)
Unable to connect with SMB1 -- no workgroup available
```

Service Analysis

Description: Review of the identified services, confirming that they match the expected functionality of an Active Directory Domain Controller.



```
(kali@kali)-[~]
$ smbclient -L //192.168.10.10 -U OLALAB\\OlaLabAdmin
Password for [OLALAB\\OlaLabAdmin]:

      Sharename      Type      Comment
-----
ADMIN$              Disk      Remote Admin
C$                   Disk      Default share
E$                   Disk      Default share
IPC$                 IPC       Remote IPC
NETLOGON             Disk      Logon server share
Share                Disk
SYSVOL               Disk      Logon server share
Users                Disk
Reconnecting with SMB1 for workgroup listing.
do_connect: Connection to 192.168.10.10 failed (Error NT_STATUS_RESOURCE_NAME_NOT_FOUND)
Unable to connect with SMB1 -- no workgroup available
```